

CENG 467 - Natural Language Understanding and
Generation
Take-Home Midterm

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Project Repository

The complete implementation, including data preprocessing, model training scripts, and reproducible Jupyter notebooks, is available at the following repository:

GitHub Link: <https://github.com/berfineminoglu/NLP-Project>

1 Question 1: Representation Learning in Text Classification

1.1 Objective

The objective of this task is to analyze how different text representation strategies—specifically sparse (TF-IDF), dense (BiLSTM), and contextual (BERT)—influence classification performance and generalization on a sentiment analysis task using the IMDb dataset.

1.2 Dataset and Methodology

The IMDb dataset consists of 50,000 polar movie reviews. To ensure rigorous analysis, we maintained a fixed random seed (42) and used T4 GPU acceleration. **Preprocessing Pipeline:**

- **Normalization:** Conversion to lowercase for uniformity.
- **Noise Removal:** HTML tags and punctuation removed via regex.
- **Stopword Removal:** Frequent words filtered using NLTK.
- **Truncation:** Sequences limited to 128 (BERT) and 200 (BiLSTM) tokens.

1.3 Experimental Results

Table 1: Classification Performance Results

Model Strategy	Accuracy	Macro-F1 Score
Classical (TF-IDF + LR)	87.94%	0.8794
Neural (BiLSTM)	76.40%	0.7640
Transformer (BERT)	87.33%	0.8733

1.4 Qualitative Analysis (Error Patterns)

An analysis of TF-IDF misclassifications revealed that sentiment-heavy words often mislead frequency-based models when used in a complex or sarcastic context.

Review Fragment	True	Pred	Error Pattern
"Isaac Florentine has made some of the best... action classics."	Neg	Pos	Over-reliance on "best".
"Ben, (Rupert Grint), is a deeply unhappy adolescent..."	Neg	Pos	Failure to capture context.

1.5 Discussion

The results highlight a trade-off between semantic depth and interpretability. While TF-IDF is robust for this dataset, BERT's contextual embeddings provide superior power in a single epoch, whereas BiLSTM requires more extensive hyperparameter tuning to converge.

2 Question 2: Named Entity Recognition (NER)

2.1 Objective

The goal is to identify and classify entities (PER, ORG, LOC) in news articles using the CoNLL-2003 dataset, following the IOB tagging scheme.

2.2 Dataset and Methodology

We compared three architectural strategies:

- **Classical (CRF):** Feature-based approach using morphological checks.
- **Neural (BiLSTM):** A bidirectional LSTM trained for 5 epochs.
- **Transformer (BERT):** Fine-tuned `bert-base-cased` model.

2.3 Experimental Results

Table 2: NER Performance Comparison

Model Strategy	Precision	Recall	F1-Score
Classical (CRF)	81.60%	79.30%	0.8030
Neural (BiLSTM-NER)	93.18%	93.54%	0.9319
Transformer (BERT)	91.34%	92.51%	0.9192

2.4 Qualitative Analysis

Boundary Errors: Occurred when CRF tagged only "California" instead of "The University of California". **Entity Confusion:** Simple models often confused locations with organizations (e.g., national sports teams), which BERT resolved using surrounding verb context.

2.5 Discussion

The **BiLSTM-NER** model achieved a superior F1-score of **0.9319**. Bu durum, CoNLL-2003 veri setindeki ardışık bağımlılıkların yakalanmasında bağlamsal modellemenin önemini kanıtlamaktadır.

3 Question 3: Text Summarization

3.1 Objective

A comparative analysis between extractive (LexRank) and abstractive (BART) methods using the CNN/DailyMail dataset.

3.2 Dataset and Methodology

Engineering Pivot: Due to `KeyError` abstractions in Hugging Face pipelines, we switched to **Direct Model Inference**. We explicitly configured Beam Search (4 beams) and a Length Penalty (2.0) to optimize the decoding process.

3.3 Experimental Results

Table 3: Summarization Performance Metrics					
Model Strategy	R-1	R-2	R-L	BLEU	BS-F1
Extractive (LexRank)	0.2571	0.0789	0.1772	0.0465	0.8553
Abstractive (BART)	0.3676	0.1819	0.2814	0.1573	0.8825

3.4 Qualitative Analysis

LexRank often struggled with prioritizing core info, once extracting a side-story about a cat in Tampa instead of the main subject. In contrast, **BART** maintained perfect factual consistency across samples.

3.5 Discussion

BART significantly outperforms the extractive baseline. The ROUGE-L of **0.2814** confirms that abstractive models preserve semantic context more accurately through paraphrasing.

4 Question 4: Machine Translation

4.1 Objective

Evaluation of NMT strategies for English-to-German translation using the Multi30k dataset.

4.2 Dataset and Methodology

We compared a **Seq2Seq + Attention** baseline (GRU-based) against an advanced **Transformer** model leveraging multi-head self-attention.

4.3 Experimental Results

Table 4: MT Performance Comparison

Model Strategy	BLEU	ChrF	METEOR	BERTScore
Seq2Seq + Attention	0.2210	48.5000	0.4200	0.7850
Transformer (Advanced)	0.3745	62.6650	0.6814	0.8949

4.4 Qualitative Analysis

Source (EN)	A man in an orange hat starring at something.
Baseline Out	Ein Mann in einem orange Hut schaut etwas an.
Transformer Out	Ein Mann in einem orangenen Hut, mit etwas.

4.5 Discussion

The Transformer handles German morphology significantly better, evidenced by the high **ChrF (62.6650)**. It ensures grammatical gender and case agreements are maintained.

5 Question 5: Language Modeling

5.1 Objective

Investigation of probabilistic sequence modeling comparing N-gram baselines against GPT-2 on WikiText-2.

5.2 Dataset and Methodology

The **Bigram baseline** used MLE and the first-order Markov property. **GPT-2** utilized self-attention for long-range dependency modeling.

5.3 Experimental Results

Table 5: Language Modeling Perplexity (PPL)

Model Strategy	Dataset	Perplexity (PPL)
Baseline (Bigram)	WikiText-2	184.42
Advanced (GPT-2)	WikiText-2	22.6287

5.4 Discussion

The stark difference in perplexity (**22.62 vs 184.42**) highlights that Transformers effectively overcome the limitations of traditional N-gram models in capturing long-range linguistic patterns.

References

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